Higher National Diploma in Engineering (Civil)
Higher National Diploma in Quantity Surveying
1st year, Second Semester Examination – 2021
ME1205 / QS 12062 - Engineering Graphics and AutoCAD – (B)

Instructions for Candidates:
Answer all questions.
You can use both sides of the paper.

No. of questions : 04
No. of pages : 03
Time : **03 hours**

06 marks allowed for completing the cage and title block.

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1)

a) A point moves such that, its distance from a fixed straight line and its distance from a fixed point both are equal to 60 mm. Draw the locus of the curve traced by that point and name it. (13 marks)

b) Draw the hyperbola through a point P which is located 30 mm and 20 mm away from the horizontal and vertical axes respectively. (13 marks)

[Total 26 marks]

2)

a) A circle of 20 mm radius is rolling on a straight line without slip. A point P is 25 mm outside from the center of the circle. Draw the locus of point P for one revolution. (Point P is initially at its lowest (bottommost) position.) (13 marks)

b) A point is moving around the surface of a cylinder of 25 mm diameter and 50 mm height, at uniform speed and at the same time it advances in the direction of axis by 35 mm (for one revolution) also with uniform speed. Draw the path of the moving point for one revolution. (13 marks)

[Total 26 marks]

3)

a) A regular hexagonal prism, edge of base 20 mm and axis 60 mm long, rests with its base on HP(Horizontal Plane) such that two of its rectangular faces are parallel to VP(Vertical Plane). It is cut by a plane perpendicular to VP, inclined at 45° to HP and passing through the right corner of the top face of the prism (A front elevation of the object is shown in Figure1). Develop the lateral surfaces of the truncated prism. (16 marks)

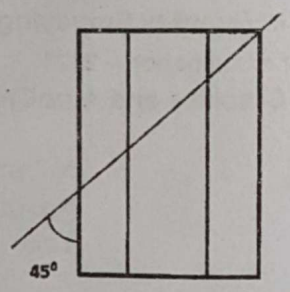

Figure 1

b) A vertical square prism, base 50 mm side is completely penetrated by a horizontal square prism, base 35 mm side, so that their axes intersect. The axis of the horizontal prism is parallel to the HP and VP, while the faces of the two prisms are equally inclined to the projection planes. (Refer to the Figure 2). Draw the lines of intersection between the two objects, in the front view, end view and plan.

(16 marks)

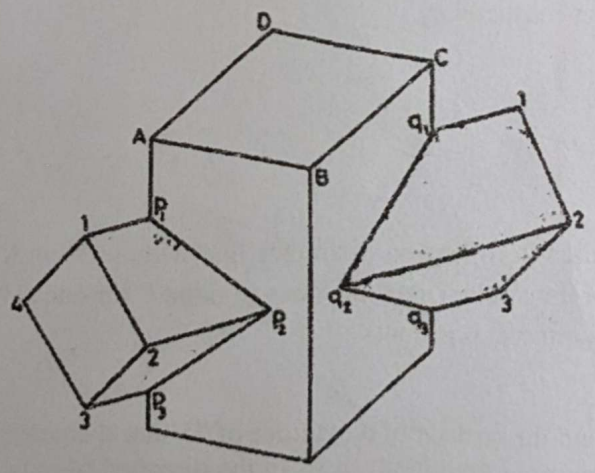

Figure 2

[Total 32 marks]

4)

Write down the necessary Auto CAD commands in order to construct the object shown in the Figure 3. All dimensions are given in millimeters. (10 marks)

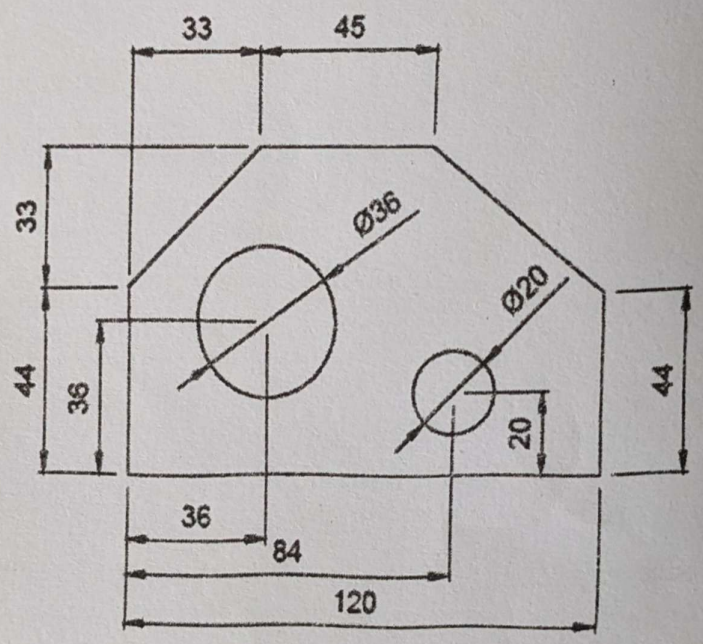

Figure 3